

Model Paper - Database Management Systems

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Q1 (20 marks)

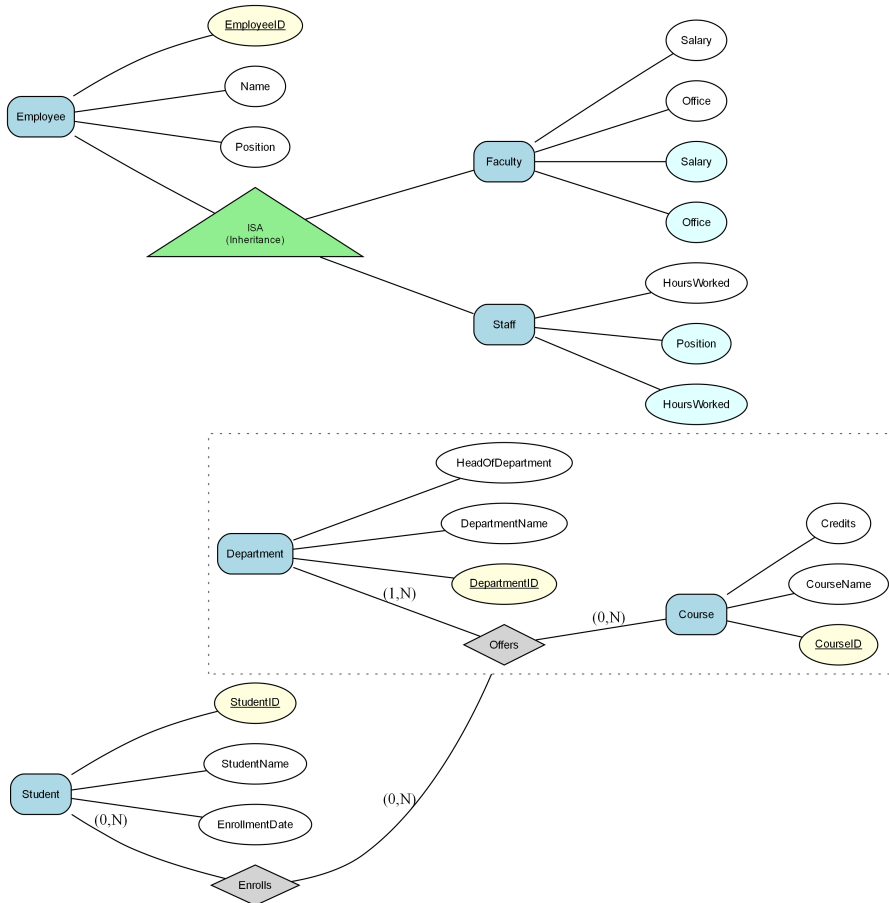
Consider the following Enhanced Entity-Relationship (EER) diagram:

Note: The diagram uses (min, max) cardinality notation where:

- (1,1) indicates one-to-one relationship (each entity participates exactly once)
- (1,N) or (1,*) indicates one-to-many relationship (one entity can relate to many)
- (0,1) indicates optional participation (zero or one)
- (0,N) or (0,*) indicates optional many participation (zero or many)

A university has multiple departments, each offering various courses. Each department has attributes such as DepartmentID, DepartmentName, and HeadOfDepartment. Courses are represented with attributes including CourseID, CourseName, and Credits. Students are enrolled in these courses, with data including StudentID, StudentName, and EnrollmentDate. The university also has an ISA hierarchy where The Employee entity has attributes such as EmployeeID, Name, Position. can be specialized into Faculty and Staff. Faculty members have attributes like Salary and Office, while Staff members have attributes like Position and HoursWorked. A Department can be related to many Courses through the 'Offers' relationship (Course participation is optional). A Student can be related to many Courses, and a Course can be related to many Students through the 'Enrolls' relationship (Student participation is optional, Course participation is optional)

Figure: EER



- Convert the following EER model into the relational model. Indicate the primary keys and the foreign keys of the resulting relations clearly. (17 marks)
- What is the best option for mapping the ISA hierarchies in the diagram? Justify your answer. (3 marks)

Q2 (15 marks)

Consider a relation $R(A, B, C, D, E)$ with the following set of functional dependencies F over R : $F = \{AB \rightarrow C, B \rightarrow D, AC \rightarrow E, DE \rightarrow A, C \rightarrow B\}$.

- Find all the keys in relation R using the attribute closure method. (4 marks)
- Is R in 3NF? Give reasons for your conclusion. (3 marks)
- Is R in BCNF? Give reasons for your conclusion. If R is not in BCNF, convert it to a set of BCNF relations. (8 marks)

Q3 (25 marks)

An e-commerce company is building a centralized database system to manage its products, customers, and orders. The schema includes tables such as `Products(product_id: integer, product_name: string, price: real)`, `Customers(customer_id:`

integer, customer_name: string, email: string), and Orders(order_id: integer, customer_id: integer, order_date: date). The system needs to ensure data integrity and proper relationships among these tables.

a) In Type 2 Driver, JDBC API calls are converted to native Java API calls. Accept or refute the above statement justifying your answer. (2 marks)

b) Code Segment: ````sql String sql = "INSERT INTO employees (id, name, age) VALUES (101, 'John Doe', 30)"; try { int rowsAffected = statement.executeUpdate(sql); if (rowsAffected > 0) { System.out.println("A new record has been inserted successfully!"); } else { System.out.println("Failed to insert a new record!"); } } catch (SQLException e) { e.printStackTrace(); } ```` Which type of statements is used in the code segment shown above? Briefly explain when this type of statement will be used. ````sql INSERT INTO Customers (customer_id, customer_name, email) VALUES (1, 'Alice', 'alice@example.com'); ```` (2 marks)

c) Briefly explain the role of foreign keys in maintaining referential integrity within the database schema. (2 marks)

d) Briefly explain how authentication and authorization are achieved in SQL Server. (3 marks)

e) An organization is establishing role-based access control for its database system. Sarah is the senior database administrator responsible for overall database administration. Emily is a junior database administrator responsible for managing the operational database. Nathan is a database developer responsible for creating and maintaining database objects. Michael is a data entry operator responsible for entering and updating transaction data.

i. Write a T-SQL statement to create a login to Sarah with windows authentication. (2 marks)

ii. Provide Sa
server using

Q4 (40 marks)

Consider the following schema of a database designed for an e-commerce platform:

Product (productId: int, name: varchar(100), price: float, stock: int) Customer

(customerId: int, fullName: varchar(100), email: varchar(50), phone: varchar(15)) Order

(orderId: int, customerId: int, orderDate: date, totalAmount: float) OrderItem (orderId: int, productId: int, quantity: int). The 'Product' table stores details about products available for purchase, including a unique product ID, name, price, and available stock. The 'Customer' table maintains customer information such as a unique customer ID, full name, email, and phone number. The 'Order' table records customer orders containing a unique order ID, customer ID, order date, and total amount of the order. The 'OrderItem' table holds information about individual items within an order, with a unique order item ID, associated order ID, product ID, and quantity of each product ordered. The 'Customer' table holds details about customers, including unique customer ID, name, email, and address. The 'Order' table manages order records, tracking customer orders with order ID, customer ID, order date, and total amount.

a) Write SQL Queries to perform the following:

i. Find the name and price of all products currently in stock. (4 marks)

ii. Find the customer ID and name of the customer who has placed the most orders.

b) b) Create a function to calculate the total amount for a given customer. This function should account for customer with a late penalty of 10% applied to those marked as 'overdue' payment status. (11 marks)

c) c) Create a trigger that automatically updates the 'totalAmount' column in the 'Order' table whenever a new order record is added, or an existing order record is updated. The 'totalAmount' column should contain the total amount for each customer. Ensure the trigger adjusts the 'totalAmount' column accurately to reflect any partial payments made by customer. The customer pays only 50% of the total amount as an initial payment when the payment status is marked as 'Partial'. (12 marks)